## Experiment 37: Correlation Between Study Time and Exam Scores

**Aim:** To identify the correlation between students' study time and exam scores using plots.

**Algorithm:**

1. Import Pandas and Matplotlib.
2. Create study time and exam score data.
3. Calculate correlation.
4. Draw a scatter plot and line plot.
5. Display the correlation value.

**Code:**

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.DataFrame({
   "Study": [1, 2, 3, 4, 5],
   "Score": [40, 50, 60, 70, 80]
})

print("Correlation:", df["Study"].corr(df["Score"]))

plt.scatter(df["Study"], df["Score"])
plt.xlabel("Study Hours")
plt.ylabel("Exam Scores")
plt.title("Study Time vs Exam Scores")
plt.show()

plt.plot(df["Study"], df["Score"], marker="o")
plt.xlabel("Study Hours")
plt.ylabel("Exam Scores")
plt.show()
```

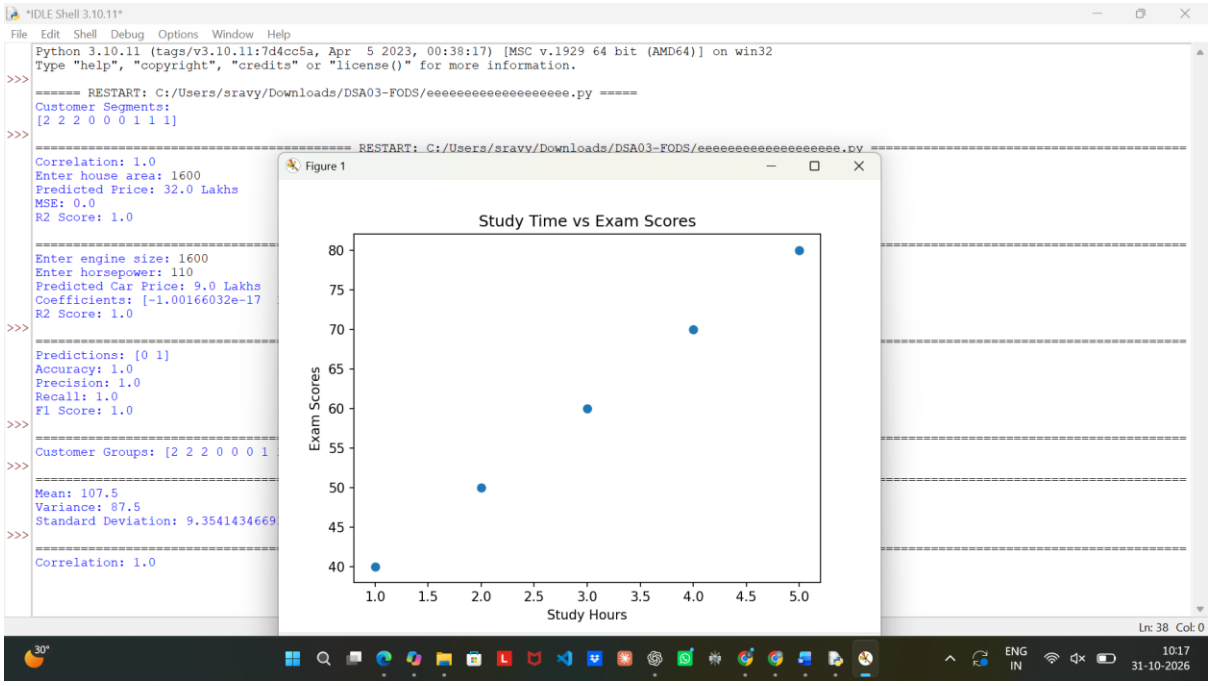
**Input:**

Study Hours: 1, 2, 3, 4, 5

Exam Scores: 40, 50, 60, 70, 80

**Output:**

Correlation: 1.0




**Result:** A positive correlation between study time and exam scores was identified and visualized.